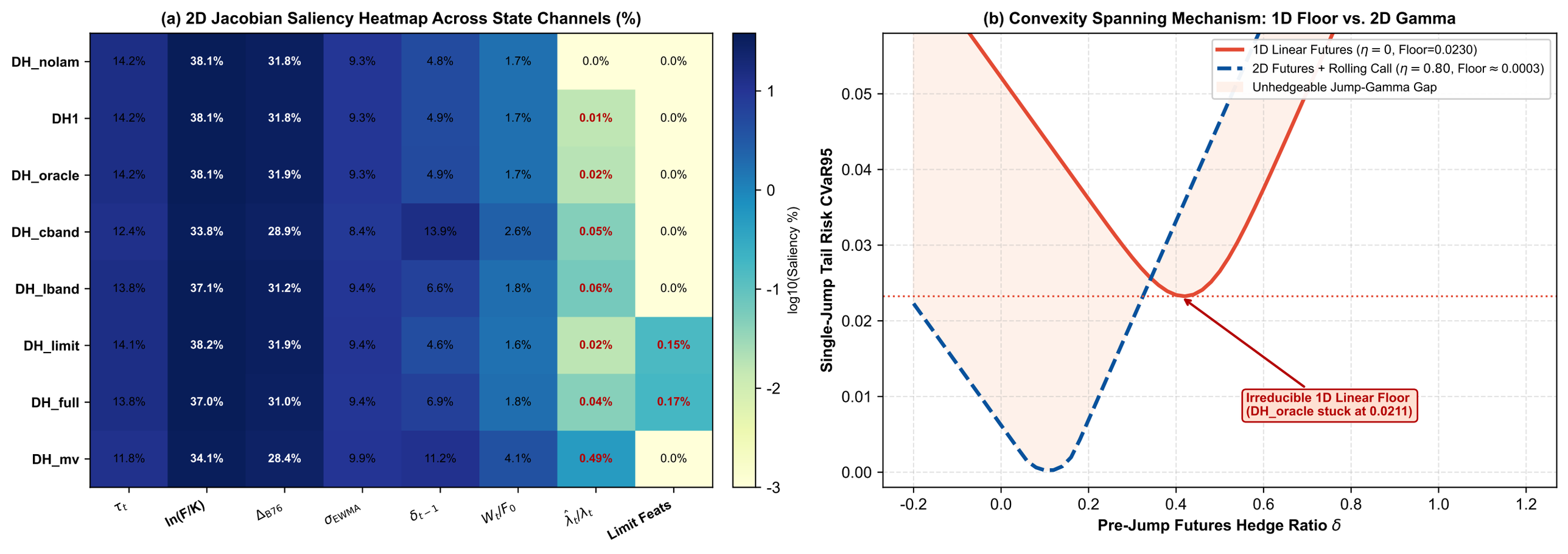

(a) 2D Jacobian Saliency Heatmap Across State Channels (%)
DH_nolam
DH1
DH_oracle
DH_cband
DH_lband
DH_limit
DH_full
DH_mv
14.2% 38.1% 31.8% 9.3% 4.8% 1.7% 0.0% 0.0%
14.2% 38.1% 31.8% 9.3% 4.9% 1.7% 0.01% 0.0%
14.2% 38.1% 31.9% 9.3% 4.9% 1.7% 0.02% 0.0%
12.4% 33.8% 28.9% 8.4% 13.9% 2.6% 0.05% 0.0%
13.8% 37.1% 31.2% 9.4% 6.6% 1.8% 0.06% 0.0%
14.1% 38.2% 31.9% 9.4% 4.6% 1.6% 0.02% 0.15%
13.8% 37.0% 31.0% 9.4% 6.9% 1.8% 0.04% 0.17%
11.8% 34.1% 28.4% 9.9% 11.2% 4.1% 0.49% 0.0%
τ_t
ln(F/K)
Δ_{B76}
σ_{EWMA}
δ_{t-1}
W_t/F_0
$\hat{\lambda}_t/\lambda_t$
Limit Feats
log10(Saliency %)
1
0
-1
-2
-3
(b) Convexity Spanning Mechanism: 1D Floor vs. 2D Gamma
1D Linear Futures ($\eta = 0$, Floor=0.0230)
2D Futures + Rolling Call ($\eta = 0.80$, Floor ≈ 0.0003)
Unhedgeable Jump-Gamma Gap
Irreducible 1D Linear Floor
(DH_oracle stuck at 0.0211)
Single-Jump Tail Risk CVaR95
0.05
0.04
0.03
0.02
0.01
0.00
-0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Pre-Jump Futures Hedge Ratio δ